

Verifiable Reasoning Framework for Legal Question Answering via RAG and Symbolic Rules

Abstract—Legal Question Answering (QA) requires the ability to correctly apply legal provisions and derive conclusions supported by explicit legal grounds. Retrieval-Augmented Generation (RAG) improves access to relevant legal documents but primarily relies on generating answers from retrieved passages. This paper proposes a framework for Vietnamese Legal Question Answering that integrates large language models (LLMs) with a structured Legal Knowledge Base and a verifiable reasoning process. Legal provisions are transformed into structured legal rules that support rule retrieval and symbolic reasoning to verify answer generation. Through experimental results on four legal domains, the proposed framework improves the consistency and reliability of legal question answering by explicitly separating knowledge representation, reasoning, and verification.

Index Terms—Legal Question Answering, Knowledge Base, Large Language Model, Retrieval-Augmented Generation, Neuro-Symbolic Reasoning

I. INTRODUCTION

Legal Question Answering (QA) plays an important role in legal consultation and decision support systems [1], [2]. Unlike general question answering, a legal answer must not only be factually correct but also be supported by appropriate legal provisions, satisfy all applicable legal conditions, and provide an explainable reasoning process [3], [4]. In practice, legal regulations are often distributed across multiple legal documents, while user queries may contain incomplete information or involve multiple legal domains simultaneously.

Recent advances in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have significantly improved language understanding and legal document retrieval [5], [6]. However, retrieving relevant legal documents alone is insufficient to guarantee legally valid conclusions if the applicable legal conditions are not systematically identified and verified before reasoning [7]. Many legal cases require integrating regulations scattered across multiple legal documents, whereas existing RAG-based systems primarily focus on document retrieval [8]. Meanwhile, neuro-symbolic reasoning provides a structured approach to knowledge representation and multi-step reasoning over heterogeneous knowledge sources [9].

This paper proposes a neuro-symbolic framework for Vietnamese Legal Question Answering, which integrates large language model (LLM), a structured Legal Knowledge Base, and a verifiable reasoning mechanism. The proposed framework explicitly separates legal knowledge representation and symbolic reasoning. It can answer a query and verify legal reasoning. Experimental results on a dataset of 443 Vietnamese legal questions spanning four legal domains using

the IRAC evaluation framework together with verification-oriented metrics. The results show that the proposed framework consistently outperforms Qwen2.5-7B-Instruct and GPT-4o, particularly on complex cases requiring reasoning across multiple legal sources.

The next section reviews related work combining RAG and symbolic rules for verification in Legal QA. Section III describes the Legal Knowledge Base serving as the framework’s symbolic layer. Section IV details the neuro-symbolic framework, combining neural understanding with symbolic reasoning for verifiable answers. Section V quantitatively evaluates the framework for Vietnamese Legal QA against baselines. Finally, the last section concludes with the main contributions of this paper and future work.

II. RELATED WORK

Neuro-symbolic reasoning combines the representation learning capability of neural networks with the interpretability and logical consistency of symbolic reasoning [10]. Representative approaches, including Markov Logic Networks (MLNs) [11], Logic Tensor Networks (LTNs) [12], [13], and DeepProbLog [14], have demonstrated the effectiveness of integrating neural learning with symbolic inference across a wide range of reasoning tasks. More recently, retrieval-enhanced frameworks such as Retrieval-Augmented Generation (RAG) [5], [15], REALM [6], and Self-RAG [16] have improved the factual reliability of large language models by incorporating external knowledge during inference. However, these approaches generally retrieve textual evidence rather than performing explicit symbolic reasoning over structured legal knowledge.

Legal QA has also attracted increasing attention in recent years. Benchmark datasets such as LexGLUE [17] and LegalBench [18] have facilitated the evaluation of legal language understanding and legal reasoning tasks. In parallel, domain-specific pre-trained language models, such as LEGAL-BERT [19], have demonstrated the effectiveness of adapting language models to legal corpora. Nevertheless, most existing Legal QA systems primarily rely on retrieving legal text passages and generating answers directly from retrieved documents, leaving the reasoning process and verification of legal conclusions largely implicit [8], [19].

Despite these advances, limited research has explored the integration of structured legal knowledge, neuro-symbolic reasoning, and explicit verification for Vietnamese Legal QA [8], [15]. It motivates the proposed framework, which represents

Vietnamese legal provisions as structured legal rules and combines neural language understanding with symbolic reasoning and verification to produce transparent, legally grounded, and traceable answers.

III. LEGAL KNOWLEDGE ORGANIZATION

This section describes the organization of the Legal Knowledge Base, which serves as the symbolic knowledge layer of the proposed neuro-symbolic framework. We first present how Vietnamese legal documents are transformed into structured legal rules, followed by the rule representation adopted for symbolic reasoning and the organization of the resulting Legal Knowledge Base to support efficient rule retrieval, reasoning, and legal traceability.

A. Legal Knowledge Base

To support symbolic legal reasoning, official Vietnamese legal documents are transformed into a structured Legal Knowledge Base composed of legal rules rather than unstructured text passages, enabling explicit symbolic reasoning and legal traceability [20]. As illustrated in Figure 1, the Legal Knowledge Base covers four legal domains while preserving the hierarchical organization of Vietnamese legal documents into chapters, articles, clauses, and points in accordance with the Law on Promulgation of Legal Documents [21].

B. Legal Knowledge Base Construction

The Legal Knowledge Base is constructed through a four-stage pipeline consisting of document preprocessing, legal structure segmentation, legal rule extraction, and knowledge base construction, as illustrated in Figure 1. The objective of this pipeline is to transform unstructured Vietnamese legal documents into structured legal rules suitable for symbolic reasoning and legal verification.

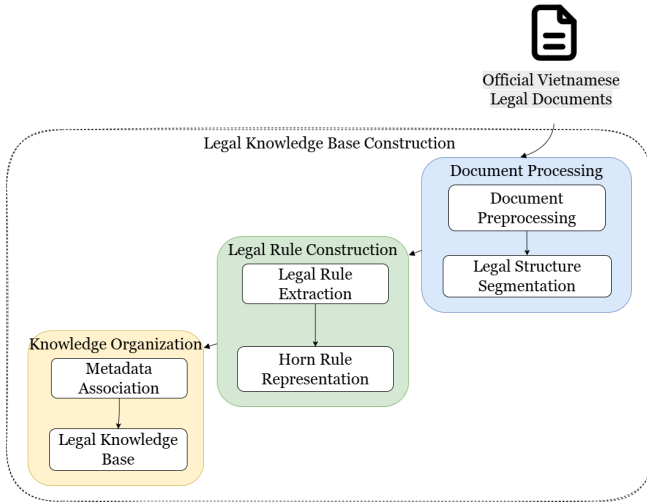


Fig. 1. Pipeline for constructing the Legal Knowledge Base.

First, legal documents are converted into plain text and preprocessed to normalize formatting while preserving their hierarchical legal structure. The documents are subsequently

segmented into chapters, articles, clauses, and points, producing legal text units for rule extraction.

From these normalized legal units, a rule-based extraction process identifies normative legal provisions and determines the corresponding legal consequence together with its prerequisite legal conditions. The legal consequence is represented as the head predicate of a Horn rule, whereas the prerequisite legal conditions are represented as body predicates. Additional legal attributes, such as the applicable subject, legal action, exceptions, and temporal constraints, are retained as structured rule attributes when applicable.

Consistent with the Horn clause representation adopted in neuro-symbolic reasoning frameworks such as Deep-ProbLog [14], each extracted legal rule is represented as

$$r_i : h_i : - b_{i1}, b_{i2}, \dots, b_{im}, \quad (1)$$

where r_i denotes the i -th legal rule, h_i represents the rule head corresponding to the legal conclusion or legal effect, and b_{ij} denotes the body predicates representing the legal conditions required for applying the rule.

For example, the legal provision requiring an enterprise to register a change of its legal representative is represented as

$$r : \text{obligation}(X, \text{register}) : - \text{enterprise}(X), \quad (2)$$

$$\text{changeRep}(X).$$

In Figure 1, each extracted rule is subsequently associated with structured metadata, including the source document, chapter, article, clause, point, legal domain, and rule identifier, to preserve legal traceability. After validation for structural correctness and completeness, the resulting rules are incorporated into the Legal Knowledge Base. The resulting Legal Knowledge Base serves as the foundation for rule retrieval, forward reasoning, legal verification, and knowledge management, while the associated metadata supports legal provenance and indexing throughout the proposed framework.

IV. NEURO-SYMBOLIC REASONING

This section presents the proposed neuro-symbolic reasoning framework for Vietnamese Legal Question Answering. Built upon the structured Legal Knowledge Base introduced in Section III, the framework combines neural language understanding with symbolic reasoning to generate legally grounded and verifiable answers.

As illustrated in Figure 2, the proposed framework consists of four sequential modules: Semantic Parsing, Candidate Rule Retrieval, Forward Reasoning, and Verification with Answer Generation. The neural component transforms a natural-language legal question into a structured legal query, while the symbolic component retrieves candidate legal rules, performs explicit forward reasoning to construct a reasoning proof, and verifies the inferred legal conclusions before generating the final legal answer. The following subsections describe each module in detail.

A. Semantic Parsing

The objective of this module is to transform a natural-language legal question into a structured legal query suitable for subsequent symbolic reasoning.

Recent advances in legal natural language understanding and retrieval-augmented reasoning employ large language models to bridge the gap between natural-language legal expressions and structured knowledge representations [5], [19]. Following this paradigm, the proposed framework uses a neural language model to extract the legal semantics required for symbolic reasoning.

The semantic parsing procedure consists of the following three steps.

Step 1. Identify the legal subject and legal action.

For natural language legal query processing, the neural language model leverages semantic context to isolate the core legal subject and action from descriptive noise. By establishing the primary legal context, these extracted elements serve as critical semantic anchors for isolating relevant case facts and driving the candidate rule retrieval module.

Step 2. Extract case facts.

Using the legal context established in the previous step, the neural language model extracts the legally relevant facts explicitly stated in the user’s question. The extraction process identifies factual information that may influence the applicability of legal rules, including personal attributes, legal status, temporal information, and quantitative information. Only factual information explicitly supported by the user’s question is retained as case facts, while assumptions or unsupported information are excluded from the structured representation. The extracted case facts constitute the initial working fact set for subsequent symbolic reasoning and are used together with the identified legal action, and legal goal during candidate rule retrieval and forward reasoning.

Step 3. Determine the target legal goal.

Based on the identified legal subject, action, and extracted facts, the neural model determines the target legal consequence as the objective for symbolic inference. It compiles these elements into a unified, structured legal query. This representation is then sent to the candidate rule retrieval module to fetch relevant rules and initialize the forward reasoning process.

B. Candidate Rule Retrieval

The objective of this module is to retrieve legal rules that are semantically relevant to the user’s legal question while reducing the search space for subsequent symbolic reasoning. Unlike conventional Legal Question Answering systems that retrieve unstructured legal text passages, the proposed framework retrieves structured legal rules from the Legal Knowledge Base, enabling symbolic inference to operate directly on explicit legal semantics.

The candidate rule retrieval procedure consists of the following two steps.

Step 1. Match legal rules.

The structured legal query generated by the semantic parsing module is matched against the legal rules stored in the Legal Knowledge Base. The matching process primarily compares the target legal goal together with the identified legal subject, legal action, and extracted case facts against the predicates of legal rules to retrieve semantically relevant candidates. The associated legal metadata, including the legal domain and document identifiers, is further used to filter and prioritize the retrieved rules while preserving legal traceability. The resulting candidate rule set may therefore contain legal rules originating from multiple legal documents or legal domains when cross-domain reasoning is required.

Step 2. Construct the candidate rule set.

The retrieved legal rules are organized into a candidate rule set for symbolic reasoning. Each candidate rule retains both its logical representation and its associated metadata [22], [23], including the source document, chapter, article, clause, legal domain, and rule identifier. By restricting symbolic inference to this candidate rule set, the framework substantially reduces the reasoning space while preserving all semantically relevant legal rules. The resulting candidate rule set is subsequently passed to the forward reasoning module, where the target legal goal serves as the stopping criterion of the symbolic inference process.

C. Forward Reasoning

The objective of this module is to derive the target legal consequence through explicit symbolic inference over the candidate legal rules. Following the classical forward reasoning paradigm over Horn clauses, legal conclusions are iteratively inferred by applying rules whose prerequisite conditions have been satisfied until no additional conclusions can be derived or the target legal goal is reached [24]. The proposed framework adopts this inference strategy to support multi-hop reasoning across legal rules and construct an explicit reasoning proof.

In Figure 3, the forward reasoning module performs iterative symbolic inference over the retrieved candidate rule set. Starting from the initial working fact set, applicable legal rules are repeatedly identified and applied to infer new legal conclusions. Each newly inferred conclusion is incorporated into the working fact set, allowing additional candidate rules to become applicable in subsequent iterations. The reasoning process continues until the target legal goal is derived or

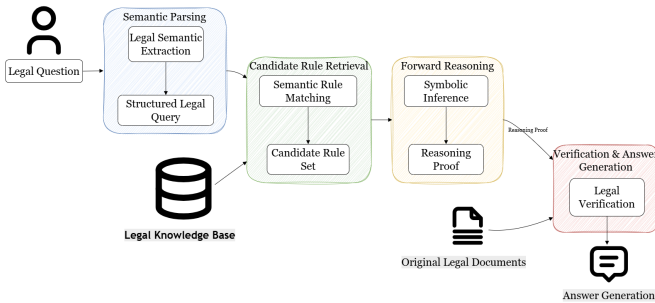


Fig. 2. Overall architecture of the proposed neuro-symbolic Legal Question Answering framework.

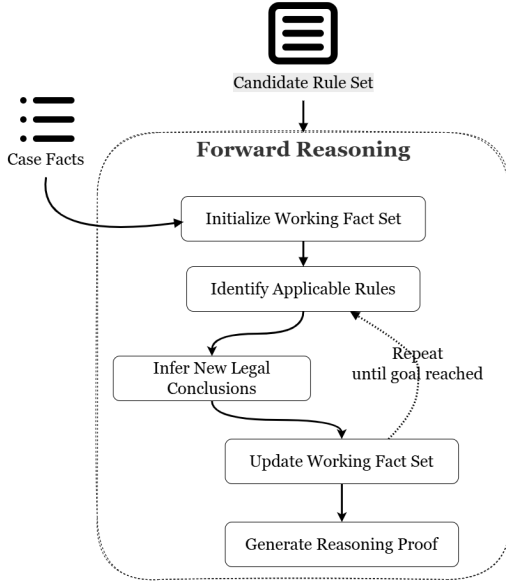


Fig. 3. Forward reasoning process for iterative symbolic inference and reasoning proof construction.

no further legal conclusions can be inferred, after which the complete reasoning proof is constructed.

The forward reasoning procedure consists of the following four steps.

Step 1. Initialize the working fact set.

The reasoning process begins by constructing the initial working fact set from the case facts extracted during semantic parsing. These case facts represent the factual circumstances of the user’s legal question and constitute the initial knowledge available for symbolic inference. Simultaneously, the candidate rule set retrieved from the Legal Knowledge Base is loaded as the symbolic reasoning space, while the target legal goal specifies the legal consequence to be derived and serves as the stopping criterion of the reasoning process. The initialized working fact set provides the basis for evaluating the applicability of candidate rules during each reasoning iteration.

Step 2. Identify applicable legal rules.

During each reasoning iteration, every legal rule contained in the candidate rule set is evaluated against the current working fact set. For each candidate rule, the framework determines whether all predicates appearing in the rule body are supported by the available case facts or by legal conclusions inferred during previous reasoning iterations. Candidate rules satisfying all prerequisite conditions are regarded as applicable and selected for inference in the current iteration, whereas the remaining rules are retained because newly inferred legal conclusions may satisfy their prerequisite conditions in subsequent iterations.

Step 3. Infer new legal conclusions.

Each applicable legal rule is applied by inferring its head predicate from the satisfied body predicates. Every newly inferred legal conclusion is immediately incorporated into the

working fact set, allowing additional candidate legal rules to become applicable during subsequent reasoning iterations. The inference process is repeatedly performed until the target legal goal is derived or no further legal conclusions can be inferred. By continuously expanding the working fact set through newly inferred legal conclusions, the framework naturally supports multi-hop reasoning across multiple legal rules and, when necessary, across different legal domains.

Step 4. Construct the reasoning proof.

After the inference process terminates, the framework organizes the complete reasoning trace into an explicit reasoning proof. For each inferred legal conclusion, the proof records the supporting case facts, the applied legal rule, intermediate legal conclusions derived during previous reasoning iterations, and the resulting legal consequence. By explicitly preserving the dependency between applicable legal rules, and inferred legal conclusions, the generated reasoning proof provides a transparent and traceable explanation of the entire symbolic reasoning process. The resulting reasoning proof is subsequently passed to the verification and answer generation module.

D. Verification and Answer Generation

The objective of this module is to validate the reasoning proof generated during forward reasoning before producing the final legal answer. Inspired by retrieval-augmented generation frameworks [5], the proposed framework retrieves the original legal documents as authoritative evidence for verification before answer generation. This additional verification layer ensures that the generated legal conclusions remain consistent with the authoritative legal sources before answer generation.

The verification and answer generation procedure consists of the following four steps.

Step 1. Identify supporting legal provisions.

The reasoning proof generated during forward reasoning is first analyzed to identify all legal provisions referenced throughout the inference process. The corresponding document identifiers, legal rule identifiers, and legal metadata are extracted to formulate retrieval queries for the original legal documents. These legal references establish the connection between the symbolic reasoning results and the authoritative legal sources used for verification.

Step 2. Retrieve the original legal documents.

Using the identified legal references, the framework retrieves the corresponding legal text, including the relevant chapters, articles, clauses, and points, from the original legal document repository. The retrieved legal text serves as the authoritative legal evidence for verification and provides an additional validation layer by comparing the symbolic reasoning results with the original legal documents.

Step 3. Verify the reasoning result.

The retrieved legal text together with the generated reasoning proof is provided to a large language model acting as a legal verifier. The verifier examines whether the legal rules applied during symbolic reasoning are supported by the retrieved legal provisions and whether the inferred legal conclusions remain consistent with the corresponding legal

text. The verification process also ensures that the final legal consequence is compatible with the authoritative legal documents without introducing unsupported legal interpretations or contradictions. After completing the consistency assessment, the verifier produces a verified reasoning result that preserves the validated reasoning proof together with the supporting legal grounds. This verified reasoning result is subsequently used as the input for natural-language answer generation.

Step 4. Generate the final legal answer.

The verified reasoning result, including the validated legal consequence and supporting legal provisions, is provided to a large language model to generate a natural-language legal response. During answer generation, the language model is instructed to preserve the verified reasoning result without introducing unsupported legal conclusions or modifying the validated legal grounds. The generated response integrates the validated legal conclusion with the corresponding supporting legal provisions into a coherent and readable explanation. Consequently, the final answer remains legally grounded, transparent, and fully traceable to the original legal documents.

V. EXPERIMENTS

This section presents the experimental setup and quantitatively evaluates the proposed neuro-symbolic framework for Vietnamese Legal Question Answering. Comparative experiments with representative large language model baselines are conducted to assess the effectiveness of the proposed approach.

A. Experimental Setup

The Legal Knowledge Base was constructed from 28 official Vietnamese legal documents covering four legal domains: social insurance, enterprise, taxation, and cross-domain legal cases. All documents were collected from official legal sources and transformed into structured legal rules following the construction pipeline described in Section III.

After preprocessing and normalization, the Legal Knowledge Base contains 28 documents, 207 chapters, 3,160 articles, 5,417 clauses, and 7,167 points. This hierarchical organization preserves the structure of Vietnamese legal documents while representing legal provisions as structured symbolic rules for legal retrieval, reasoning, and verification. Consequently, every legal rule in the candidate rule set remains traceable to its original legal provision, forming the symbolic knowledge layer of the proposed framework.

The proposed framework is compared with Qwen2.5-7B-Instruct and GPT-4o under identical experimental settings using deterministic decoding ($temperature = 0.0$, $top-p = 1.0$) and the same prompting strategy.

Answer quality is evaluated using the IRAC framework, a widely adopted method for assessing legal reasoning and legal argumentation. Inspired by recent evaluations of large language models on legal reasoning tasks [18], [25], each answer is evaluated across five dimensions: Issue (I), Rule (R), Analysis (A), Conclusion (C), and Legal Grounding (L). Each dimension is independently scored on a scale from 0 to 1 according to a predefined evaluation rubric, and an overall

IRAC score is assigned based on these evaluation dimensions. All scoring results are subsequently reviewed by a legal expert to improve evaluation reliability.

B. Experimental Results

The evaluation dataset consists of 443 Vietnamese legal questions collected from real-world legal consultation scenarios across four legal domains. Each question is paired with a reference answer and the corresponding legal grounds. The quantitative comparison across legal domains is summarized in Table I.

TABLE I
IRAC PERFORMANCE ACROSS LEGAL DOMAINS

Domain	Cases	Qwen	GPT-4o	Our method
Social Insurance	196	50.66	60.68	80.92
Enterprise	119	50.87	59.01	71.22
Taxation	69	49.64	61.90	68.88
Cross-domain	59	38.01	47.43	77.82
Overall	443	48.87	58.66	76.03

Table I and Figure 4 demonstrate that the proposed framework consistently outperforms both baselines across all domains, achieving the highest overall IRAC score of 76.03. The largest gain occurs on Cross-domain questions, where it substantially exceeds GPT-4o and Qwen2.5-7B-Instruct, indicating that explicit symbolic reasoning benefits multi-provision integration. Among single-domain categories, Social Insurance yields the highest performance, while Enterprise and Taxation remain more challenging. This suggests that diverse legal conditions and multiple regulations present greater reasoning difficulty, even with structured knowledge.

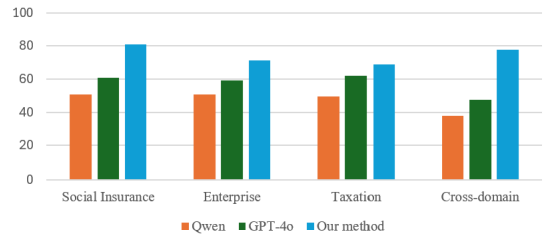


Fig. 4. Comparison between the systems on each domain.

While GPT-4o generated fluent, plausible answers and Qwen2.5 handled straightforward questions. Both baselines frequently omitted essential legal conditions, cited incomplete grounds, or drew premature conclusions. These limitations became particularly evident for Cross-domain questions requiring legal knowledge from multiple regulations.

The proposed framework addresses these limitations by explicitly separating legal rule retrieval, symbolic reasoning, verification, and answer generation. Instead of relying on implicit language model reasoning, legal conclusions are derived only after satisfying all required conditions during forward reasoning. Consequently, every generated answer is supported by explicit legal grounds and a traceable process,

enhancing consistency and reliability. These findings suggest the framework’s primary contribution lies not in stronger language generation, but in augmenting language models with explicit legal reasoning capabilities and legal traceability.

VI. CONCLUSION AND FUTURE WORK

This paper presents a neuro-symbolic framework for Vietnamese Legal Question Answering that integrates a structured Legal Knowledge Base, LLMs, and a verification-supported reasoning process. By representing legal provisions as structured legal rules and explicitly separating knowledge representation and symbolic reasoning, the proposed framework provides more transparent and reliable legal reasoning than conventional retrieval-based approaches.

Experimental results on a dataset of 443 Vietnamese legal questions demonstrated that the proposed framework consistently outperformed general-purpose large language models, particularly on complex legal questions requiring multi-step reasoning across multiple legal provisions. The verification mechanism further improved the reliability of generated answers by ensuring consistency between the reasoning process and supporting legal grounds.

Future work will focus on extending the Legal Knowledge Base to additional legal domains, enhancing multi-step reasoning capabilities, and developing automatic methods for updating and maintaining legal knowledge as regulations evolve. These improvements will further increase the scalability and practical applicability of the proposed framework for large-scale Legal Question Answering systems.

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